

VB-101 Computational Discovery and Prioritization of a Cardiac Regeneration Therapeutic

1. Project title

Cross-Species Single-Cell and Spatial Multi-Omic Identification of Therapeutically Actionable Cardiomyocyte Regeneration Programs in Human Myocardial Injury

Program

VB-101 — Myocardial Regeneration Gene Therapy Vector

Development stage

Discovery / computational preclinical research

2. Executive summary

VB-101 will establish a computational framework for identifying and prioritizing therapeutic regulators capable of promoting cardiac regeneration in diseased adult myocardium.

The project will not begin by selecting a predetermined gene such as YAP, Agrin, ERBB2, MYC, PTMA, or another established regenerative factor. Instead, it will first define the **cellular state associated with productive cardiac regeneration**, determine which components of that state are conserved across regenerative organisms and developmental windows, determine how that program is altered in adult human myocardial injury, and then computationally identify upstream regulatory perturbations predicted to move diseased human cardiomyocytes toward the regenerative state.

The central conceptual model is:

regenerative biology → conserved cardiomyocyte state → human disease deficit → regulatory control nodes → virtual therapeutic perturbation → VB-101 lead

The analysis will integrate:

- single-cell RNA sequencing
- single-nucleus RNA sequencing
- single-cell ATAC-seq
- spatial transcriptomics
- temporal regeneration datasets
- cross-species orthology

- regulatory-network inference
- cell-cell communication
- computational perturbation
- cardiac functional/electrophysiological gene programs

The primary output will be a ranked set of **1–3 therapeutically actionable targets**, with one lead VB-101 candidate.

Importantly, the computational study will establish a **testable therapeutic hypothesis**, not claim that regeneration, electrical integration, efficacy, or safety has been experimentally demonstrated.

3. Scientific premise

Adult mammalian cardiomyocytes have extremely limited regenerative capacity, whereas regenerative competence is retained during defined developmental windows and in organisms such as zebrafish.

Single-cell studies have demonstrated that regenerative responses involve distinct cardiomyocyte states rather than simply a generalized increase in proliferation. Neonatal mouse studies have identified injury-responsive immature cardiomyocytes with distinct transcriptional and regulatory programs.

A separate neonatal mouse study combined scRNA-seq and scATAC-seq and demonstrated cell-type-specific regulatory networks and extracellular mediators associated with cardiomyocyte proliferation, angiogenesis and regeneration.

Recent work further demonstrates that regeneration involves the surrounding niche. For example, cardiac fibroblast-derived versican promotes cardiomyocyte proliferation through integrin $\beta 1$ /ERK1/2/Akt signalling, while macrophage metabolic state can influence regenerative proliferation.

YAP-associated adult renewal has also been shown to generate a neonatal-like cardiomyocyte state accompanied by specific fibroblast and macrophage populations, reinforcing the concept that regeneration is a multicellular state rather than a CM-autonomous proliferation switch.

Finally, recent work directly connects regenerative versus non-regenerative repair to restoration versus disruption of the cardiac conduction system.

These findings motivate a computational strategy that simultaneously considers:

CM state + niche + regulatory network + spatial context + functional identity.

4. Central hypothesis

Primary hypothesis

Adult human diseased cardiomyocytes retain a partially accessible but suppressed subset of regulatory programs associated with regenerative cardiomyocyte states.

A therapeutically useful VB-101 target should be capable of increasing the regenerative-state program while preserving cardiomyocyte functional identity and avoiding pathological proliferation, fibrosis and electrical instability.

Formally:

$$\textit{Human diseased CM} \rightarrow \textit{Regenerative CM state}$$

should be computationally achievable through a small number of regulatory perturbations.

5. Primary research question

Which regulatory perturbation is predicted to move a viable adult human myocardial-injury cardiomyocyte toward a conserved regenerative state while preserving functional/electrophysiological identity and minimizing pathological programs?

6. Secondary research questions

RQ1

What transcriptional and regulatory state characterizes productive cardiac regeneration?

RQ2

Which components of this state are conserved across zebrafish, mouse, pig and human biology?

RQ3

Which regenerative programs are lost during postnatal mammalian maturation?

RQ4

Which regenerative programs remain partially accessible in human myocardial injury?

RQ5

Which upstream regulators control the largest fraction of the regenerative state?

RQ6

Are candidate regulators present in the relevant human cardiomyocyte populations and anatomical regions?

RQ7

Are their regulatory loci accessible in diseased human cardiomyocytes?

RQ8

Would perturbing them preserve mature cardiac functional identity?

RQ9

Which candidate provides the best predicted therapeutic benefit/risk ratio?

7. Experimental philosophy

The project will follow six rules.

Rule 1 — Regeneration $\neq$ proliferation

Ki-67, PCNA, TOP2A or cell-cycle activation alone will never constitute evidence of regeneration.

Rule 2 — Cells $\neq$ biological replicates

Inference will respect donor/animal/sample structure.

Rule 3 — Discovery $\neq$ validation

Datasets used to construct the model will not be used to claim independent validation.

Rule 4 — Expression $\neq$ causality

Candidates must have regulatory/mechanistic evidence.

Rule 5 — Human relevance is a gate

A target that works only in an animal model will not become a VB-101 lead.

Rule 6 — Computational prediction $\neq$ therapeutic proof

The project ends with a prioritized hypothesis requiring experimental validation.

8. Dataset architecture

Tier I — Regeneration discovery

A. Zebrafish regeneration

PRJNA1233465

Use for:

- temporal regeneration
- CM state transitions
- regenerative versus non-regenerative phases
- spatial organization
- multicellular interactions

This is particularly valuable because the study provides both single-cell and spatially resolved regeneration data across multiple post-injury stages.

B. Neonatal mouse regeneration

GSE130699 / Cui et al.

Use for:

- regenerative CM states
- proliferative immature CM populations
- transition out of regenerative competence
- mammalian regulatory programs

The underlying study identified distinct CM populations across healthy, injured and regenerating hearts and identified transcriptional programs associated with regenerative immature cardiomyocytes.

C. Neonatal mouse multi-omic regeneration

Wang et al. / PMID 33296652

Use for:

- scRNA-seq
- scATAC-seq
- regulatory networks
- extracellular mediators
- cell-cell signalling

This dataset is especially important for connecting expression programs to regulatory DNA.

Tier II — Large-mammal translation

D. Pig regenerative/non-regenerative model

GSE185289

Use for:

- fetal versus postnatal states
- regenerative versus non-regenerative injury
- large-mammal conservation
- translational filtering

This is a major bridge between rodent regeneration and human disease.

E. Independent pig validation

GSE293028

Use as a held-out dataset.

Primary purpose:

determine whether the regenerative program discovered from other datasets reproduces in an independent large-mammal dataset.

9. Tier III — Human therapeutic evaluation

F. Kuppe human myocardial infarction atlas

Use as the principal human disease dataset.

Required modalities:

- snRNA-seq
- snATAC-seq
- spatial transcriptomics

Required anatomical comparisons:

- infarct
- border zone
- remote myocardium

This dataset is particularly important because it allows the candidate to be evaluated simultaneously at the:

RNA + chromatin + spatial levels.

G. Heart Cell Atlas v2

Use as the primary healthy human reference.

Applications:

- healthy CM baseline
- cardiac-cell specificity
- conduction-system comparison
- non-CM off-target assessment
- regulatory comparison

The presence of conduction-system populations is especially valuable for the VB-101 electrical-safety analysis.

H. GSE217494

Use as independent human disease validation.

Conditions include:

- healthy
- acute MI
- chronic ischemic cardiomyopathy
- non-ischemic cardiomyopathy

The purpose is not to discover the candidate, but to determine whether the candidate's human disease association reproduces independently.

10. Data inclusion criteria

Every dataset must pass a formal eligibility checklist.

Required

- cardiac tissue
- appropriate injury/regeneration phenotype
- usable biological-replicate metadata
- identifiable cardiomyocyte population
- accessible condition/time metadata
- public accession
- adequate sample-level information

Preferred

- multiple biological replicates
- temporal sampling
- spatial information
- multimodal data
- independent validation cohort

Exclude

- bulk-only studies when single-cell resolution is required
- datasets with irrecoverable condition metadata
- datasets where samples cannot be mapped confidently to experimental groups
- cell lines as evidence for in vivo regeneration
- datasets with unresolved sample identity
- studies where the regenerative phenotype cannot be distinguished from generic injury

If metadata are ambiguous:

HALT.

No inference from accession names or biological intuition.

11. Phase I — Dataset audit and harmonization

For every sample, construct:

```
dataset
species
study
sample
animal/donor
technology
injury
control
time
anatomical region
sex
age
replicate
cell type
sequencing modality
```

Create a version-controlled metadata manifest.

No downstream analysis begins until the metadata audit passes.

12. Phase II — Quality control

Perform modality-specific QC.

scRNA/snRNA

Assess:

- genes detected
- counts
- mitochondrial fraction where appropriate
- ribosomal fraction
- ambient RNA
- doublets
- low-quality nuclei
- sample-specific abnormalities

scATAC

Assess:

- fragment counts
- TSS enrichment
- nucleosomal pattern
- FRiP
- doublets
- library complexity

Spatial

Assess:

- tissue quality
- spot/cell coverage
- sequencing depth
- spatially restricted artifacts

QC thresholds will be dataset-specific but documented before biological comparisons.

13. Phase III — Cell identity

Initial annotations will use:

1. original study annotations
2. curated markers
3. reference mapping
4. independent annotation checks

Primary populations:

- cardiomyocytes
- fibroblasts
- endothelial cells
- smooth muscle/pericytes
- macrophages
- other immune cells
- epicardial cells

Cardiomyocytes will subsequently be reclustered independently.

14. Phase IV — Cardiomyocyte state atlas

Within cardiomyocytes, identify:

- mature/quiescent CM
- immature CM
- injured/stressed CM
- hypoxic CM
- regenerative CM
- proliferative CM
- hypertrophic CM
- failing CM
- apoptotic/dying CM
- redifferentiating CM where supported

The distinction between:

proliferative

and

regenerative

will be explicit.

15. Phase V — Build the Regenerative Cardiomyocyte State

Construct the **VB-101 Regenerative Cardiomyocyte Signature (RCS)**.

It will contain several independently derived modules.

Module R1

Regenerative/developmental plasticity

Module R2

Controlled cell-cycle activation

Module R3

Cytokinesis/division competence

Module R4

Metabolic remodeling

Module R5

ECM/mechanical adaptation

Module R6

Pro-survival/repair

Module F1

Sarcomere/contractile preservation

Module F2

Calcium handling

Module F3

Electrical/conduction identity

Module P1

Fibrosis

Module P2

Pathological inflammation

Module P3

Hypertrophy

Module P4

Uncontrolled proliferation

The final state is therefore multidimensional.

16. Regenerative State Score

Define:

$$RSS = R_{regen} + R_{cycle} + R_{metabolic} + R_{repair} + R_{ECM} - P_{pathology}$$

and separately:

$$FIS = F_{sarcomere} + F_{calcium} + F_{electrical}$$

The principal therapeutic objective becomes:

$$\Delta RSS > 0$$

while:

$$\Delta FIS \geq 0$$

and:

$$\Delta Pathology < threshold$$

17. Phase VI — Cross-species conservation

Map only high-confidence orthologs across:

- zebrafish
- mouse
- pig
- human

For each gene/module calculate:

- direction consistency
- effect size
- temporal consistency
- CM specificity
- regenerative-state association

A candidate will receive a **Conservation Score**.

A gene found only in zebrafish but not mammalian regeneration is not automatically discarded, but it cannot receive the same confidence as a conserved mammalian regulator.

18. Phase VII — Developmental loss-of-regeneration analysis

Explicitly model:

Regenerative neonatal CM → *mature adult CM*

Identify:

Which regulatory programs disappear as regenerative competence is lost?

This creates a highly important candidate class:

"Regeneration-suppressed regulators"

These are potentially more therapeutically interesting than genes that merely become activated after injury.

19. Phase VIII — Injury versus regeneration discrimination

For every candidate, compare:

Regeneration

against:

Injury

Stress

Inflammation

Fibrosis

Pathological proliferation

Candidates that simply respond to injury will be penalized.

20. Phase IX — Regulatory-network inference

Use:

pySCENIC

to infer transcription-factor/regulon activity.

decoupleR

for independent TF/pathway activity inference.

Coexpression modules

for data-driven network structure.

PPI networks

for mechanistic connectivity.

Construct a consensus regulatory network.

Candidates will be prioritized as upstream regulators rather than downstream biomarkers.

21. Phase X — Dynamic state transitions

Use:

CellRank

to infer state-transition probabilities.

moscot

to model temporal population-level transitions.

The principal question:

Injured CM → Regenerative CM → Functional/recovered CM

versus:

Injured CM → Stress/fibrosis/failure

Candidate regulators will be associated with the transition rather than merely the endpoint.

22. Phase XI — Human disease mapping

Freeze the regenerative model.

Then expose it to the human datasets.

Determine:

1. Regenerative score

How far are human diseased CMs from the regenerative state?

2. Module preservation

Which regenerative modules remain?

3. Suppression

Which modules are lost?

4. Cellular specificity

Which CM population carries the deficit?

5. Anatomical localization

Infarct vs border vs remote.

The central translational hypothesis will be tested here.

23. Phase XII — Spatial confirmation

Use **cell2location** to map CM states and candidate activity into human spatial data.

The critical question:

Is the therapeutic target located in the biologically relevant myocardium?

Prioritize candidates associated with:

- viable border-zone CM
- injured but surviving CM
- repair-competent CM

and avoid candidates whose apparent signal originates predominantly from:

- fibroblasts
- macrophages
- endothelial cells
- necrotic tissue

unless the candidate is deliberately a niche-targeting therapeutic.

24. Phase XIII — Multicellular regenerative niche

The project will explicitly test whether the candidate is:

CM-autonomous

or

niche-dependent.

Use:

LIANA+

for broad cell-cell communication analysis.

Then:

NicheNet

for ligand-to-target mechanistic prioritization.

This allows two therapeutic architectures to emerge:

Architecture A

Direct CM manipulation.

Architecture B

Manipulation of a regenerative niche signal that acts on CMs.

The data determine which is superior.

25. Phase XIV — Human chromatin feasibility

For candidates with human disease relevance, interrogate scATAC-seq.

Determine:

- promoter accessibility
- enhancer accessibility
- disease-state accessibility
- CM-specific accessibility
- transcription-factor binding potential

A candidate whose RNA is absent because its regulatory locus is completely inaccessible may require a fundamentally different therapeutic strategy than one whose expression is merely suppressed.

This distinction is therapeutically important.

26. Phase XV — Electrical identity safeguard

Build an independent **Cardiac Functional Identity Score**.

Evaluate:

Electrical

- ion channels
- sodium handling
- potassium currents
- calcium channels
- conduction-related genes
- gap junctions

Calcium

- calcium cycling
- SR function
- excitation-contraction coupling

Structural

- sarcomere
- contractile machinery

A candidate is penalized if predicted regeneration comes at the cost of a major loss of mature cardiac identity.

Recent experimental work showing that regenerative and non-regenerative hearts differ in restoration versus disruption of conduction networks makes this a necessary component rather than an optional safety analysis.

27. Phase XVI — Virtual perturbation

For every high-ranking regulator:

$$Target \uparrow$$

and/or

$$Target \downarrow$$

depending on the biological direction.

Propagate the perturbation through the regulatory network.

Calculate:

$$\Delta RSS$$

$$\Delta FIS$$

$$\Delta Pathology$$

and:

$$Therapeutic\ Utility = \Delta RSS + \Delta FIS - \Delta Pathology$$

The result is a predicted state transition rather than a single-gene effect.

28. Minimum-sufficient perturbation analysis

Evaluate:

single targets

$$A$$

combinations

$$A + B$$

$$A + C$$

$$B + C$$

and, only where justified:

$$A + B + C$$

The optimization objective is:

achieve the regenerative phenotype with the smallest intervention.

This is preferable to automatically designing a multi-gene therapy.

29. Benchmarking

The system must recover established biology.

Known pathways/factors will function as benchmark controls, including:

- YAP/Hippo biology
- Agrin-associated repair
- neonatal CM regulatory programs
- ECM-mediated regeneration
- macrophage-mediated regenerative signalling

For example, YAP-associated renewal has been shown to involve a neonatal-like CM state and a multicellular fibroblast/macrophage niche.

Agrin/ECM-associated biology and newer factors such as PTMA and HMGB2 provide additional contemporary benchmarks; PTMA has recently been reported to promote CM proliferation and regeneration and HMGB2 has been implicated in a metabolic regenerative axis.

The framework must be able to rediscover established biology before novel candidates are considered credible.

30. Negative controls

Include:

Random matched genes

Matched for:

- expression

- gene length
- detection
- cell specificity

Injury genes

Should not automatically rank as regenerative.

Generic proliferation genes

Should fail if they damage functional identity.

Fibrotic genes

Should score poorly.

Non-cardiac genes

Should fail the specificity gate.

31. Statistical validation

The central statistical unit will be the:

animal/donor/sample

rather than the individual cell.

Use:

- pseudobulk models
- mixed-effects models where appropriate
- sample-aware differential expression
- differential abundance
- bootstrap confidence intervals
- permutation testing
- multiple-testing correction

Sensitivity analyses will determine whether conclusions depend on:

- integration method
- cell-state definition
- gene-set definition
- species
- individual dataset

- weighting scheme
-

32. Leave-one-study-out validation

This is a major robustness test.

Repeatedly perform:

Train on $N - 1$ datasets

and:

Test on the held – out dataset

For example:

- leave out one zebrafish study
- leave out one mouse study
- leave out one pig study

The regenerative score should remain predictive.

This prevents the model from becoming a disguised signature of one laboratory.

33. Leave-one-species-out validation

An even stronger test:

Train without pig.

Ask:

Does the resulting regenerative program predict pig regeneration?

Then train without mouse.

Ask:

Does it generalize to mouse?

This tests whether the conserved program is genuinely cross-species.

34. Human external validation

Human disease data will be separated into:

Development cohort

Used to evaluate candidate relevance.

Independent validation cohort

GSE217494.

A candidate that disappears in the second cohort is downgraded.

35. Candidate ranking

Every candidate receives scores for:

Domain	Score
Regenerative-state contribution	0–1
Cross-species conservation	0–1
Human disease relevance	0–1
CM specificity	0–1
Border-zone relevance	0–1
Regulatory evidence	0–1
Chromatin feasibility	0–1
Dynamic-state association	0–1
Functional preservation	0–1
Therapeutic actionability	0–1
Vector feasibility	0–1
Pathological proliferation risk	0–1
Fibrotic risk	0–1
Electrical risk	0–1

The final ranking will be generated using a prespecified weighting system plus sensitivity analysis.

36. VB-101 lead-selection gates

A candidate cannot become the lead unless it passes:

Gate 1 — Biological conservation

Evidence in ≥ 2 regenerative systems.

Gate 2 — Independent replication

Reproduces outside the discovery dataset.

Gate 3 — Human relevance

Present/relevant in human diseased myocardium.

Gate 4 — CM relevance

Signal is localized to an appropriate cardiomyocyte population.

Gate 5 — Mechanistic evidence

Candidate occupies an upstream regulatory position.

Gate 6 — State-transition prediction

Perturbation increases regenerative-state probability.

Gate 7 — Functional preservation

No major predicted collapse of mature cardiac identity.

Gate 8 — Safety

No unacceptable predicted pathological program.

Gate 9 — Robustness

Ranking survives sensitivity analyses.

Gate 10 — Therapeutic feasibility

A realistic intervention direction exists.

37. Expected candidate classes

The analysis could ultimately produce one of several outcomes.

Outcome A — Single-gene regulator

A single upstream regulator dominates.

Outcome B — Two-node combination

No single gene adequately restores the state, but two regulators do.

Outcome C — CM + niche intervention

Regeneration requires both CM intrinsic and extracellular signalling.

Outcome D — State restoration without proliferation

The dominant therapeutic opportunity may be functional recovery rather than proliferation.

Outcome E — No sufficiently safe target

This is a valid result.

The project must be allowed to conclude:

no candidate satisfies the predefined criteria.

That protects against confirmation bias.

38. Software environment

Core

Python

scRNA/snRNA

- Scanpy
- AnnData
- scvi-tools

Statistical/compositional

- Pertpy
- pseudobulk modelling
- edgeR/DESeq2-compatible statistical framework

Dynamics

- CellRank
- moscot

Regulatory

- pySCENIC
- decoupleR

Spatial

- cell2location

Cell-cell communication

- LIANA+
- NicheNet

Chromatin

- Signac
- ArchR

Reproducibility

- Git
- Snakemake
- Docker/Conda
- pinned package versions

39. Why the architecture is intentionally redundant

No single computational method will determine the VB-101 target.

Confidence should increase when independent evidence converges:

Expression

+

Chromatin

+

Regulatory activity

+

Trajectory

+

Spatial localization

+

Cross – species conservation

+

Human disease relevance

+

Virtual perturbation

A candidate supported by all eight dimensions is substantially more compelling than one supported by differential expression alone.

40. Primary endpoint

The primary computational endpoint is:

Change in the validated Regenerative State Score of human diseased cardiomyocytes following a predicted candidate perturbation, subject to preservation of the Functional Identity Score and predefined pathology constraints.

41. Secondary endpoints

1. Cross-species regenerative-state conservation.
2. Human MI regenerative-state deficit.
3. Candidate CM specificity.

4. Spatial enrichment in relevant myocardial regions.
 5. Regulatory-network centrality.
 6. Chromatin accessibility.
 7. Predicted cell-state transition probability.
 8. Cell-cell signalling relevance.
 9. Electrophysiological-risk score.
 10. Robustness to dataset/model perturbation.
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42. Deliverables

Deliverable 1

VB-101 Cardiac Regeneration Atlas

Cross-species, sample-aware atlas.

Deliverable 2

VB-101 Regenerative Cardiomyocyte Signature

Frozen and independently validated.

Deliverable 3

Regenerative Regulatory Network

Ranked upstream regulators.

Deliverable 4

Human MI Regeneration Deficit Map

Cellular + spatial + regulatory.

Deliverable 5

Virtual Perturbation Matrix

Candidate × predicted phenotype.

Deliverable 6

VB-101 Therapeutic Ranking

Top candidates with uncertainty.

Deliverable 7

Lead Candidate Development Dossier

For the lead:

- target
 - direction of perturbation
 - target cell
 - mechanism
 - evidence
 - human relevance
 - spatial localization
 - chromatin feasibility
 - functional preservation
 - predicted safety liabilities
 - vector feasibility
 - experimental validation requirements
-

43. Experimental handoff

The computational project ends at:

"Candidate X is the highest-confidence therapeutic hypothesis."

It does not claim:

"Candidate X regenerates human myocardium."

The experimental program would then test:

Level 1

Molecular perturbation.

Level 2

CM proliferation/cytokinesis.

Level 3

New myocardium / lineage evidence.

Level 4

Scar reduction and cardiac function.

Level 5

Electrical integration.

Level 6

Arrhythmia.

Level 7

Biodistribution and off-target effects.

Level 8

Large-animal efficacy and safety.

44. Falsification criteria

The hypothesis is considered unsupported if:

1. no reproducible regenerative CM state exists across independent datasets;
 2. the signature fails cross-species validation;
 3. human diseased myocardium does not exhibit the predicted deficit;
 4. candidates are not reproducibly associated with regulatory control;
 5. virtual perturbations cannot restore the state;
 6. predicted restoration consistently causes functional-identity loss;
 7. all high-ranking candidates fail safety criteria;
 8. results are driven primarily by batch/species rather than biology.
-

45. Success criteria

The project succeeds if it demonstrates:

<i>Conserved regenerative state</i>

plus:

Human disease relevance

plus:

Mechanistically plausible regulatory control

plus:

Robust virtual therapeutic rescue

plus:

Preserved cardiac functional identity

and produces a small number of experimentally testable therapeutic hypotheses.

46. Final conceptual model

The complete VB-101 computational experiment is:

Regenerative organisms

↓

Regenerative cardiomyocyte state

↓

Cross – species conserved modules

↓

Upstream regulatory network

↓

Human MI/HF state deficit

↓

Spatial + chromatin confirmation

↓

Virtual perturbation

↓

Regeneration + Functional preservation – Pathology

↓

VB-101 Lead

47. What makes this different from a conventional cardiac transcriptomics study

The project is not:

"Find genes upregulated after cardiac injury."

It is not:

"Find genes enriched in neonatal hearts."

It is not:

"Train a classifier distinguishing regenerative and non-regenerative cells."

It is not:

"Predict that YAP is important."

It is:

Infer a conserved regenerative cellular state, identify the regulatory machinery controlling that state, determine which components remain therapeutically accessible in human diseased myocardium, and computationally identify the smallest perturbation predicted to restore regenerative competence without sacrificing cardiac function.

That is the appropriate computational discovery program for VB-101.